HEALTHCARE USE CASE · RELIABLE, CLEAN BACKUP POWER

Keeping hospitals powered when the grid fails

Focus region Nigeria, then sub-Saharan Africa | **Team solution** GridRescue AI

MINI-CASE STUDY

Hauwa runs the power plant room of a 150-bed hospital in Kano. Most nights the grid goes off for hours. Each time, the theatre and the intensive-care unit go dark for half a minute while the diesel generator starts, and nurses restart machines by hand. This year the price of diesel more than doubled in four months. Three solar companies have sent quotes that look nothing alike, and the hospital board has asked her one question: which one is right for us?

Persona is illustrative, drawn from documented conditions in Nigerian hospitals.

WHY IT MATTERS

When the power goes, the first people hurt are patients on breathing machines, mothers in labour, newborns in incubators and people on dialysis. Only half of hospitals in sub-Saharan Africa have reliable electricity [1]. In Nigeria, more than one person in three still has no electricity at all [2], and about 4 in 10 working primary health centres have no power [3].

WHAT WE ARE SOLVING FOR: THE THREE A'S

ACCESS	AFFORDABILITY	AVAILABILITY
Bring reliable power to communities and regions still living without it.	Drive down the cost of clean energy so it is within reach for everyone.	Build resilient systems that keep energy flowing anytime, anywhere.
Today: about 4 in 10 health centres in Nigeria have no power [3].	Today: diesel went from ₦1,362 to ₦3,277 a litre in four months of 2026 [4].	Today: the national grid collapsed 12 times in 2024 [5].
We help: any hospital or clinic can be checked in seconds, for free, from open data.	We help: we pick the right size — not too big, not too small — so no money is wasted.	We help: life-saving machines never switch off, even in a full-day blackout.

THE PROBLEM AT A GLANCE

WHAT IS HAPPENING	WHAT IT MEANS
The national grid collapsed 12 times in 2024 — about once a month [5].	Backup must last through the whole night, not just a few minutes.
Diesel more than doubled in price in early 2026 [4]; one teaching hospital reported energy bills of over ₦100 million a month [6].	Running on diesel is becoming unaffordable.

WHAT IS HAPPENING	WHAT IT MEANS
About 22 million generators keep Nigerian homes and businesses running [7].	Generators are the default — the alternative must be cheaper and more reliable.
Safety rules say life-saving power must be back within 10 seconds [8]; generators often take longer.	Hospitals need instant backup, not only a generator.
Many past solar projects failed because they were the wrong size and had no money for maintenance [9].	Getting the size right, and planning for upkeep, matters as much as the panels.

OUR SOLUTION — GRIDRESCUE AI

GridRescue AI is a free online advisor. A hospital drops a pin on a map and describes itself. In seconds, our AI reads the site — sunshine, wind, rivers, and how often the grid fails there — tests eight different backup designs hour by hour through a power cut, and gives a clear answer: what to install, why, what it costs and when it pays for itself.

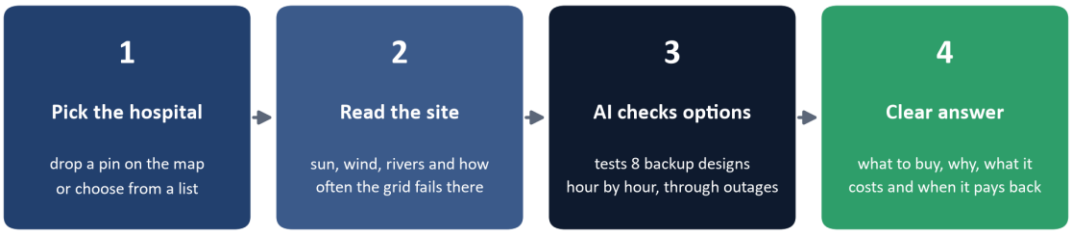


Figure 1. How GridRescue AI works, in four steps.

The answer always follows the same simple plan, so the hospital never goes dark:



Figure 2. Three layers of protection: instant backup for life-saving machines, solar and a Backup Device for the night, and the generator only as a last resort.

HOW THE AI DECIDES

Our AI works like a careful engineer. First it makes a quick, educated guess, using what it learned from 8,000 example sites across Nigeria and Africa; its guess matches the full check about three times out of four. Then it double-checks the guess with a full hour-by-hour test, and tries the hard cases:

- a whole-day national blackout
- the cloudiest month of the year

- a power cut that lasts longer than planned
- diesel prices jumping again

If any test fails, the AI says exactly what to add, then explains its decision step by step in plain words. Hospitals see every step on the screen, so they can trust the answer.

EXAMPLE: A 150-BED HOSPITAL IN KANO

This is what GridRescue AI recommends for the hospital in our story:

QUESTION	ANSWER
What should they install?	Instant backup for the operating theatres and intensive care, solar panels covering about the area of a football pitch (roof and grounds), and a large Backup Device (lithium iron phosphate, LFP).
Will the lights stay on?	Yes. Every essential service stays on through a 12-hour night power cut — and through a full-day national blackout.
How much diesel does it save?	About 247,000 litres a year.
What does it cost?	About US$ 1.5 million — and it pays for itself in about 3.5 years from diesel and electricity savings.

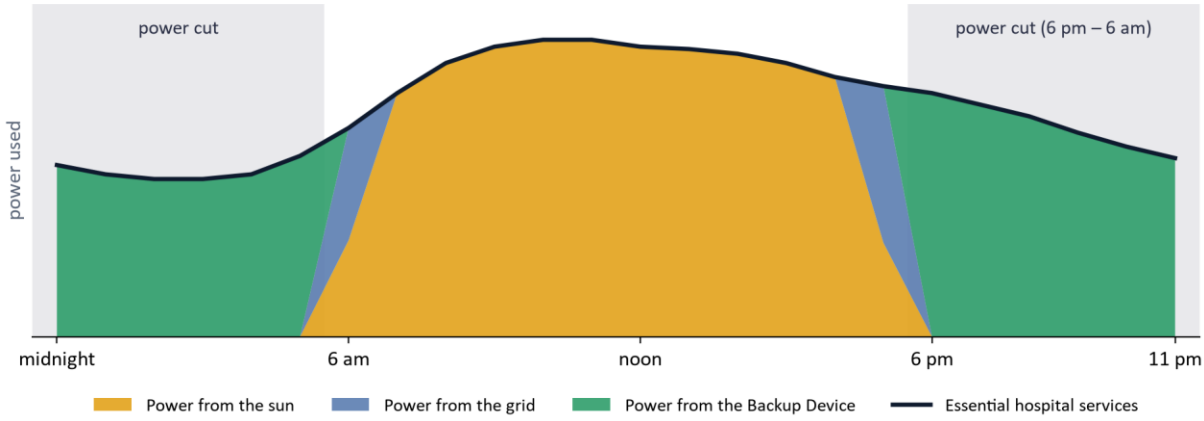


Figure 3. One day at the Kano hospital. The sun powers the hospital by day and charges the Backup Device; the Backup Device carries every essential service through the night power cut — with no diesel. Our solar estimate is checked against EU satellite data for Kano [10].

HOW WE MEET THE JUDGING CRITERIA

JUDGED ON	OUR ANSWER
Impact	Life-saving power never stops. One hospital saves about 247,000 litres of diesel and about 1,090 tonnes of CO_2 a year.
Technical feasibility	Built and working today on real public data; solar, Backup Devices and backup units are proven, and getting cheaper [11]. Every result is tested hour by hour.

JUDGED ON	OUR ANSWER
Scalability	Free, online, no special software or accounts. Works for a small village clinic or a large teaching hospital, anywhere in Nigeria and Africa.
Community fit	Written for hospital boards, not engineers: plain words, costs in naira and dollars, and a clear next step.

ONE-PAGE IMPACT ESTIMATE

Results from GridRescue AI for two typical facilities in Kano. Real sites will differ.

	150-BED HOSPITAL	SMALL HEALTH CENTRE
Essential services kept on in a power cut	Yes — all night	Yes — all night
Diesel saved per year	≈ 247,000 litres	≈ 13,000 litres
CO_2 avoided per year	≈ 1,090 tonnes	≈ 60 tonnes
Cost to install	≈ US$ 1.5 million	≈ US$ 74,000
Pays for itself in	≈ 3.5 years	≈ 3 years

At scale: 100 hospitals and 1,000 health centres would save about 38 million litres of diesel every year. Nigeria's health programmes already target nearly 20,000 facilities [3].

OUR TEAM SUBMITS A working prototype, a 3-minute pitch and this impact estimate — judged on impact, technical feasibility, scalability and community fit.

REFERENCES

[1] World Health Organization, "Electricity in health-care facilities," Fact sheet, Aug. 31, 2023. [Online]. Available: https://www.who.int/news-room/fact-sheets/detail/electricity-in-health-care-facilities

[2] World Bank, "Access to electricity (% of population) — Nigeria," World Development Indicators, EG.ELC.ACCS.ZS. [Online]. Available: https://data.worldbank.org/indicator/EG.ELC.ACCS.ZS?locations=NG

[3] Sustainable Energy for All, "A data-driven approach to health facility electrification in Nigeria," Mar. 29, 2022. [Online]. Available: https://www.seforall.org/news/a-data-driven-approach-to-health-facility-electrification-in-nigeria

[4] D. Ibemere, "NBS: Diesel price rises 86.40% to N3,277/litre in May 2026," Legit.ng, Jun. 24, 2026. [Online]. Available: https://www.legit.ng/business-economy/energy/1716140-nbs-diesel-price-rises-8640-n3277litre-2026/

[5] P. Uzoho, "Nigeria's power grid collapses again, 12th time this year," ThisDay, Dec. 11, 2024. [Online]. Available: https://www.thisdaylive.com/index.php/2024/12/11/nigerias-power-grid-collapses-again-12th-time-this-year

[6] J. Emmason, "Electricity tariff: Life-saving services threatened in teaching hospitals," Leadership, Jun. 2024. [Online]. Available: https://leadership.ng/electricity-tariff-life-saving-services-threatened-in-teaching-hospitals/

[7] World Bank, "Nigeria Distributed Access through Renewable Energy Scale-up Project (P179687): Project Information Document," Rep. PIDC35402, Dec. 2022. [Online]. Available: https://documents1.worldbank.org/curated/en/099160503012330661/pdf/P1796870167a360d0b0da016429be36e28.pdf

[8] P. E. Mackey, "Hospital emergency power supply systems," Consulting-Specifying Engineer, Jul. 20, 2022. [Online]. Available: https://www.csemag.com/hospital-emergency-power-supply-systems/

[9] A. Al-Akori, "PV systems for rural health facilities in developing areas: A completion of lessons learned," IEA PVPS, Rep. T9-15:2014, Nov. 2014. [Online]. Available: https://iea-pvps.org/wp-content/uploads/2020/01/IEA-PVPS_T9-15_2014_PV_for_rural_health_facilities.pdf

[10] European Commission JRC, "PVGIS 5.3 solar estimate for Kano (12.00° N, 8.52° E)," queried Oct. 10, 2026. [Online]. Available: https://re.jrc.ec.europa.eu/api/v5_3/PVcalc?lat=12.0&lon=8.52&peakpower=1&loss=14&outputformat=json

[11] International Renewable Energy Agency, Renewable Power Generation Costs in 2025: Executive Summary. Abu Dhabi, UAE: IRENA, Jul. 2026. [Online]. Available: https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2026/Jul/IRENA_TEC_RPGC_2025_Executive_summary_2026.pdf

Sources accessed 10 October 2026. All figures are early estimates to be confirmed with local quotes.